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Framing Education as a Complex Service System: A Structured and Multidimensional Framework Through the Service Design Lens

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Abstract

This article explores the application of service design principles to higher education, positioning universities as complex service systems that extend beyond the academic realm. The goal is to propose a structured framework highlighting service design's role in enhancing pedagogical practices, optimizing support systems, and fostering collaborations between universities and society. Through a comparative analysis of existing frameworks, the study identifies commonalities and gaps, creating a positioning matrix that visually represents the relationships. The research identifies four key quadrants: (i) Learning Services, (ii) Services for Learning, (iii) Services for Educational Infrastructure, and (iv) Educational Interaction Services. The findings emphasize that recognizing higher education as a complex service system is essential for understanding its broader influence on economic development, innovation, and social justice. We also highlight that the proposed matrix can be a starting point for educators, administrators, policymakers, and service design professionals to apply service design concepts more effectively in educational contexts.

Keywords: Service Design, Higher Education, Complex Systems.

Introduction

The design field has progressed over the decades, reflecting significant contextual and theoretical shifts. It has transitioned from a product-centered perspective to an

approach that recognizes the complexity and interconnectedness of social and ecological systems (Ceschin & Gaziulusoy, 2016). Similarly, service design has evolved significantly over the years, moving from viewing services in terms of their intrinsic differences from products to a broader and more holistic approach that emphasizes the co-creation of value and the interconnectedness of various stakeholders (Morelli et al., 2021).

With its multidisciplinary approach, service design focuses on creating and enhancing services by integrating various perspectives and methodologies (Patrício et al., 2018; Prestes Joly et al., 2019) and has the potential to drive organizational change (Junginger & Sangiorgi, 2009). It has proven to be a versatile approach, applicable across various fields, and has improved quality and efficiency in sectors such as healthcare, transportation, and education (Ding et al., 2023; Musolino et al., 2022; Palazzo et al., 2024).

In the context of education, service design can generally be understood in two distinct categories: as a specific course focus on teaching service design as an academic discipline (Becermen & Simeone, 2019) or as a transversal approach, where service design principles are applied to improve the overall educational experience (Blizzard, 2020; Ostrom et al., 2011). In the first category, service design is an independent course or discipline with a specialized curriculum that prepares students for service design careers. These programs emerged in the 1990s, became more common in universities, and are offered at various levels, including bachelor's, master's, and specialized diplomas (Mager et al., 2023). This approach can be referred to as "service design in education."

On the other hand, when viewed as a transversal approach, service design can be integrated into existing curricula and practices, shaping the development of overall educational systems and experiences. The focus is on viewing educational processes as service systems and engaging all stakeholders—students, faculty, administrators, and external entities like communities and companies—in co-creating the value of education. This perspective can be referred to as "service design for education." Both perspectives are related to the education field but differ in their application. One is more academic, focusing on the discipline itself, while the other is more practical, influencing how education is experienced and delivered.

While educational institutions are widely recognized as essential institutions in society, they remain underexplored when viewed through the lens of service systems (Lella et al., 2012), and the potential of service design principles in this field has not yet been broadly recognized (Ding et al., 2023). One possible reason for struggling to understand educational institutions through a service lens is the challenge of defining

education itself. Over the years, many concepts about education have been outlined, and these concepts vary according to the social, economic, and political context of the period in which they are analysed. Harford & O'Donoghue (2020) argue that education can be defined by a pattern or norm that gives it purpose, where moral values are necessarily implicitly present. That means that education's moral values are thus profoundly embedded in how universities interact with students, communities, and society, making the concept of education and the functioning of educational institutions complex and multifaceted.

This complexity reveals that the current educational structure is both a product of historical and cultural practices and a cause of ongoing (or not) societal and environmental challenges (Morin, 1999). In this way, education systems must critically examine the principles that underlie their functioning, creating a dynamic process of re-education that redefines values and reshapes the system from within. Jayapalan (2005) points out that education should be capable of harmonizing with the changing world, aligning itself with the rapid developments taking place in various fields. Recognizing this dynamic nature reinforces the importance of systemic approaches that avoid fixed or reductionist definitions of what education is or should be.

In this context, the transversal approach to service design can play a crucial role in facilitating transformational change within educational institutions (Kuzmina & Bhamra, 2014), especially when this transformation requires questioning and changing the fundamental assumptions, beliefs, norms, and values that support the organizational structure (Junginger & Sangiorgi, 2009; Morelli et al., 2021). However, to effectively drive this change, it is essential to establish a clear understanding of the field where transformation is intended. Effective representation and communication are vital for fostering collaboration among designers and diverse stakeholders, ensuring the development of systemic solutions (Morelli & Tollestrup, 2007). Therefore, this research proposes a framework for positioning higher education as a complex service system through the service design lens. The goal is to offer a structured, multidimensional approach that emphasizes the role of service design in enhancing pedagogical practices, optimizing support systems, and fostering meaningful collaborations between universities and their broader societal contexts. This framework aims to illustrate how education, when viewed as a complex service system, can be more effectively aligned with the needs of its stakeholders while promoting systemic improvements that contribute to institutional growth and societal development.

This research employs an exploratory approach of comparative analysis of existing frameworks to identify commonalities and differences (Livari et al., 2000),

understanding various methodologies and their interrelationships. As this analysis is preliminary, the focus will be on three frameworks, with which the authors are already familiar, by taking three steps: (i) review of existing frameworks based on the authors' prior knowledge of relevant models; (ii) identification of commonalities and complementary features, involving a qualitative comparison of key components across the selected frameworks and understanding how each framework addresses service design and which gaps can be filled by combining the three; (iii) development of the positioning matrix, which aims to visually represent the relationships between the frameworks and highlight areas for further exploration.

Educational Perspectives: Education through Mechanistic, Marketized, and Complex Lenses

Throughout history, education has been shaped by various approaches that reflect different views on the role of learning, the function of schools, and the interactions between individuals in the educational process (Harford & O'Donoghue, 2020). In this theoretical background, we will discuss three perspectives that can be observed to understand how education is conceived and practiced: (i) mechanistic, (ii) marketized, and (iii) complex. Each of these perspectives offers a distinct interpretation of education and, consequently, influences teaching methodologies, the relationships between students and professors, and the role of educational institutions in society.

The mechanistic perspective, inspired by the Industrial Revolution and Cartesian philosophy, views education as a linear and systematic process where teaching and learning occur in a rigid, fragmented, and standardized manner, similar to the operation of a machine (Barbosa, 2010). This perspective advocates for a structured education system that maximizes efficiency, employing techniques such as division of labor and specialization (Batista & Salvi, 2006; Guedes, 2013; Menegazzi, 2019). In this context, the teaching process is not seen as dynamic or interactive but rather as a series of standardized steps to prepare students to meet the demands of the labor market (Pinheiro & Herrero, 2018) or society efficiently and predictably.

On the other hand, the marketized perspective is linked to transforming, especially higher education institutions (HEIs), into market-driven entities that adopt traditional business practices such as branding, marketing, and quality control to become competitive in a global market. This process has led to adopting the "student-as-client" metaphor, where the student is seen as a consumer of educational services, similar to a customer in any commercial market (Díaz-Méndez et al., 2019). This view

simplifies the complexity of educational services and leads universities to prioritize "customer" (student) satisfaction rather than focusing on the quality of education and the critical development of students.

While the mechanistic and marketized perspectives simplify the complexity of educational services, the complex systems perspective offers a more nuanced understanding, emphasizing education's dynamic, interconnected, and adaptive nature. According to Snowden & Boone (2007) and Zhou (2012), complex systems can be characterized by six key features: (i) they involve a large number of elements; (ii) they involve a large number of relationships between these elements; (iii) these relationships are nonlinear, and small changes can lead to significant consequences; (iv) the system is dynamic, and the whole is greater than the sum of its parts; (v) solutions cannot be imposed but emerge from circumstances (the principle of emergence); and (vi) the characteristics of these elements and relationships exhibit some degree of uncertainty or imprecision. In other words, the complex view acknowledges that interactions between parts are neither static nor constant but rather a dynamic, nonlinear, multicausal, and fuzzy (ambiguous) process.

When applied to education, this complex perspective offers a more dynamic and integrated view of the learning process. It recognizes education as an adaptive, interdependent system where students, professors, institutions, and society interact nonlinearly (Jacobson et al., 2019). As such, education must be approached as a flexible, responsive, and continuously evolving system, with value co-created through the interactions of all involved. This view contrasts with more mechanistic and marketized models, proposing a richer, more interconnected understanding of how educational systems function and evolve.

In this sense, education from the complexity perspective requires a new outlook on educational practices, prioritizing flexibility, adaptation, and value co-creation, aligning with education through the lens of service. Under this perspective, if the educational process is viewed as a collaborative, co-created experience where different actors in the educational system, such as students, professors, institutions, and the community, collaborate, then service design principles can be a valuable approach to address challenges and enhance the educational system (Blizzard, 2020). In this way, learning is not viewed as a unilateral transaction but as a dynamic process involving the use and application of resources by all involved (Díaz-Méndez et al., 2019). Education, therefore, becomes a service system in which value is co-created, not merely consumed.

Understanding Service Design for Higher Education

The concept of service design for higher education presents an innovative approach to improving educational systems by applying its principles, methods, and tools. Rather than focusing only on traditional learning processes, service design can enhance every aspect of the educational experience, including teaching, learning, and the support services that sustain students, faculty, and society (Blizzard, 2020; Faust, 2011; Lella et al., 2012).

In this way, Lella et al. (2012) explore universities as service systems that operate from two interconnected perspectives: external and internal. From the external perspective, universities are seen as entities that significantly impact their communities, contributing to local economic development, generating jobs, and providing knowledge and talent. This interaction with the community highlights the importance of universities as engines of innovation and support for the regional economy, as well as emphasizing the mutual dependence between institutions and various stakeholders, such as students, businesses, and government agencies. On the other hand, the internal perspective focuses on the organizational structure of universities, analysing their functions and operational domains, such as education, research, and administration. This internal analysis seeks to understand how different parts of the university collaborate to provide quality services and how they can improve value co-creation among internal stakeholders, such as students and faculty.

Another key characteristic of education systems is their significant variation across regions and countries, shaped by distinct policies and conditions that influence the development and implementation of educational guidelines. Therefore, any changes within these systems must be understood in the context of each specific educational environment (Kuzmina & Bhamra, 2014). In this context, by using service design principles, universities can better respond to the needs of their internal and external environments, driving innovation and improving service delivery across all dimensions of the academic and administrative system while always respecting the local context.

From another perspective, Faust (2011) discusses how universities should address their responsibilities to better meet students' needs by offering two main types of services: (i) Learning Services, which focus on the quality of teaching and direct support for student learning (such as the development of teaching methodologies and provision of pedagogical resources), and (ii) Services for Learning, which provide broader support services (such as access to libraries, enrolment services, and administrative assistance) to help students navigate the academic infrastructure of

the university. Faust's central argument is that to be effective, universities must offer both types of services. By integrating these services into a cohesive system, universities can create a more user-centered, efficient, and responsive educational environment, better support students' academic and non-academic needs (Blizzard, 2020) and aligning with the core principles of service design.

In a broader view, Morelli et al. (2021) position services within three distinct logical levels, each with its focus and importance in value creation. The first level, "service as interaction," concentrates on the dynamics of value co-creation that occur during direct interactions between service beneficiaries and other actors, highlighting the significance of the experiences lived at that moment. In the educational field, this level can focus on the dynamics of value co-creation during direct interactions between students, faculty, and other stakeholders, emphasizing the significance of the lived experiences during these interactions, which directly influence students' learning and engagement.

The second level, "service as infrastructure," addresses the organization and design of the processes and spaces that enable these interactions, emphasizing how the infrastructure should be designed to facilitate value creation. It highlights the importance of designing educational spaces and systems—both physical and digital—to facilitate value creation and enhance student experience. Finally, the third level, "service as a systemic institution," considers the broader context in which services operate, including cultural, social, and political factors that shape service practices and value co-creation. This multi-level approach offers a comprehensive understanding of how service design can be integrated into the educational system, ensuring that every level interacts and complements the others.

Thus, service design for education provides an opportunity to rethink and improve educational systems by promoting collaboration, enhancing the experiences of all stakeholders, and creating a more adaptive and sustainable environment. Improving and transforming educational systems through service design include reconfiguring students as active members of the academic community rather than passive consumers (Díaz-Méndez et al., 2019), prioritizing incremental and localized changes (Kuzmina, 2014) that are easier to manage and deliver faster results in the first moment, and using service design principles to guide policy-making and resource allocation, ensuring alignment with institutional priorities (Blizzard, 2020). Blizzard (2020) also advocates using service design to facilitate organizational change by involving staff and stakeholders in co-creating solutions and strengthening informal networks within institutions to enhance collaboration.

By focusing on collective value creation, service design moves away from traditional, top-down educational models, paving the way for a more dynamic, sustainable, and engaging educational experience.

Positioning Matrix for Higher Education as a Complex Service System

The selection of Faust (2011), Lella et al. (2012), and Morelli et al. (2021) as foundational references for this framework is both strategic and complementary. Each of these authors contributes a distinct but interconnected perspective to the conceptualization of education as a complex service system. Faust offers a micro-level view, focusing on internal pedagogical and infrastructural services within the university, thus clarifying the service logic embedded in learning and institutional support. Lella et al. enrich the meso-level understanding by proposing a dual perspective (internal and external) on universities as service systems, which enables the examination of institutional interdependencies. Morelli et al. extend the analysis to the macro level, positioning services across three logical layers: interaction, infrastructure, and systemic institutions.

While each of these frameworks offers valuable insights at different levels, this paper integrates them into a unified positioning matrix that enables a systemic reconfiguration of educational services. The matrix, illustrated in Figure 1, organizes educational services into four quadrants formed by two intersecting axes: Internal versus External Perspectives, and Service as Interaction versus Service as Infrastructure. This integration establishes a multidimensional foundation for understanding higher education through the lens of service design, offering a strategic tool to visualize, analyze, and redesign services in alignment with the inherent complexity of multilayered institutional systems.

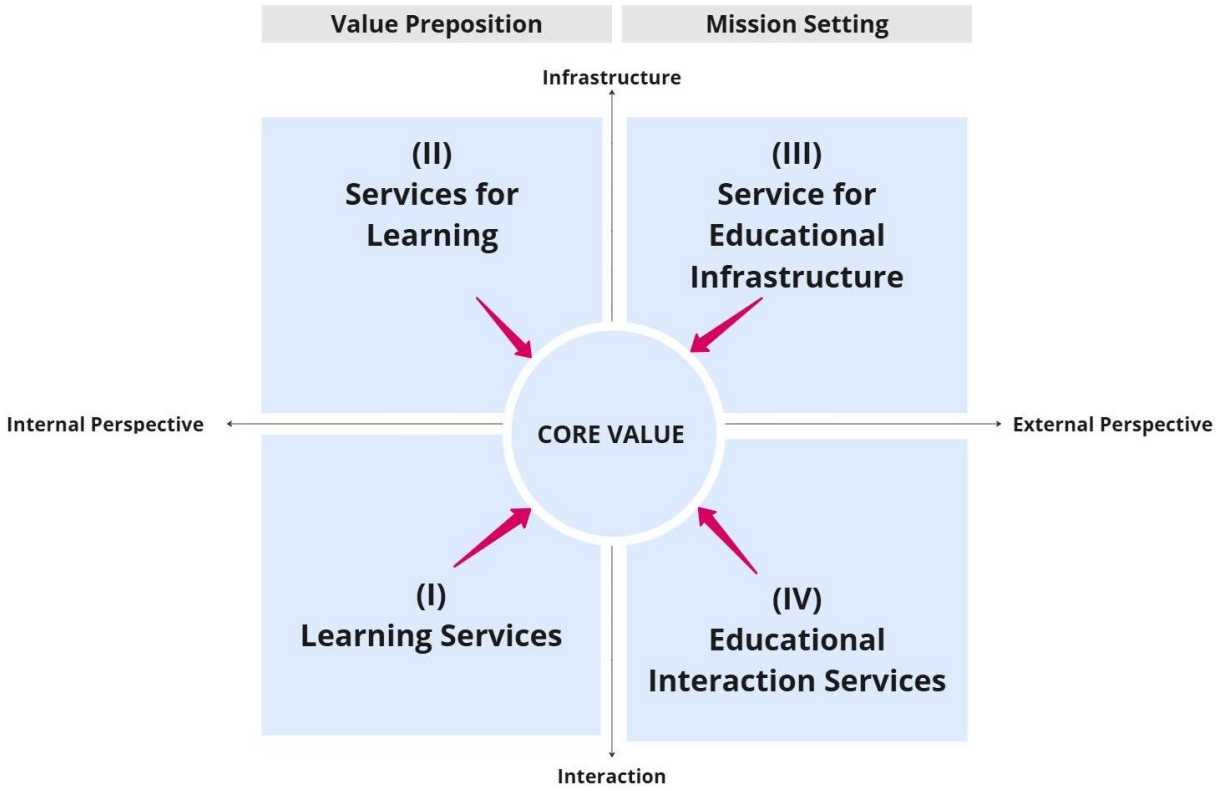


Figure 1. Positioning Matrix for Higher Education as a Complex Service System

The first quadrant (internal + interaction) corresponds to Learning Services and explores service design as a pedagogical approach within the university's internal context, focusing on how professors can become designers of learning experiences. Professors can actively create instructional materials and learning environments (Calavia et al., 2023) that enhance the educational process by applying service design methodologies, such as user-centered design, co-creation of value, and iterative improvement. This category also included all processes that directly support these activities, such as curriculum design, teaching methods, lecture preparation, professor's training, and the provision of educational resources.

The second quadrant, Service for Learning (internal + infrastructure), addresses how service design can optimize internal services and infrastructure to better support the university's educational objectives. Here, the focus is on the internal systems that support internal and non-academic operations. This quadrant ensures that the internal infrastructure is designed to facilitate the educational mission by improving administrative processes, academic support systems, campus facilities, IT support, and library management. These first two quadrants align closely with the concept of a value proposition (Vargo & Lusch, 2004), emphasizing the co-creation of value

between the university and its students. In both cases, the value is not simply transferred to the student but co-created through interactions, collaboration, and optimizing the resources that support the learning experience. In this case, the university makes value propositions by designing and facilitating environments that allow students to co-create value, rather than simply delivering a one-way service.

The third quadrant, Service for Educational Infrastructure (external + infrastructure), highlights the role of external systems and policies in supporting the educational process. It focuses on aligning government policies, accreditation procedures, funding mechanisms, community partnerships, and technological infrastructure to create a robust ecosystem for effective learning. This perspective operates at the macro level, considering broader societal and cultural factors that shape the educational system. These external influences, including government policies, educational reforms, global trends, and evolving societal values, determine how education is perceived, funded, and implemented.

The last quadrant (external + interaction), which can be seen as an Educational Interaction Service, emphasizes services that improve the interactions between the educational institution and external stakeholders, enhancing communication, cooperation, and mutual understanding between the university and its surrounding community. It highlights service design education as an opportunity both for students to apply their learning to a real-life context and for external actors (local communities, companies, local public administration) to experiment or generate social innovation; this may happen, for instance, through collaboration with external stakeholders in projects, workshops, or other training or research activities that address societal challenges, offering students valuable experiential learning opportunities and contributing to meaningful social impact.

The third and fourth quadrants are categorized as "mission setting" once they focus on how the external environment shapes and supports educational institutions' missions. The third quadrant focuses on aligning infrastructure and external systems to support the institution’s mission, while the fourth quadrant ensures that the institution’s mission is connected to real-world challenges and societal needs, creating a reciprocal relationship between the institution and its external stakeholders. These quadrants reinforce the institution’s overall purpose and can guide its operational and strategic direction.

Together, these four quadrants offer a comprehensive framework for understanding the educational field as a complex service system. Each quadrant focuses on a specific aspect of the system—from direct student interactions to broader institutional

and societal impacts. Table 1 organizes and synthesizes the key aspects of each category.

	Value Proposition Services		Mission Setting	
Aspect	**(I) Learning Services**	**(II) Services for Learning**	**(III) Service for Educational Infrastructure**	**(IV) Educational Interaction Services**
Definition	Services that generate value propositions for the internal improvement of teaching and learning processes	Services that generate value propositions by optimizing internal infrastructure processes to support learning	Services that connect educational institutions to the social and cultural context related to external macro-level policies	Services that improve interactions between educational institutions and local stakeholders external to the university
Focus Area	Internal teaching and learning processes	University infrastructure and operations	External agencies environment	External communities, public and private stakeholders
Key Stakeholders	Professors, students, program coordinators, pedagogical advisors, academic support professionals	Administrative staff (library, enrolment, housing, IT services), students, professors	Policymakers, government agencies, global educational organizations, program coordinators	Professors, students, program coordinators, pedagogical advisors, local communities, local companies, public administration, research stakeholders

Table 1. The Key Aspects of the Positioning Matrix

In positioning higher education as a complex service system, the proposed matrix offers a practical and strategic tool to guide multiple stakeholders in identifying, analysing, and improving the services they design, manage, or experience within academic ecosystems. Far from being a merely conceptual model, the matrix provides actionable insights for both internal and external actors, who can use it to reflect on their roles, responsibilities, and opportunities for value co-creation.

For students, the matrix helps clarify the systemic nature of their educational journey, showing how their learning experience is shaped not only by direct pedagogical interactions but also by invisible service structures such as administrative processes, campus infrastructure, and institutional partnerships. When engaged in participatory initiatives, students can use the matrix to map pain points and suggest improvements across all quadrants, becoming active agents in shaping educational quality. On the other hand, professors, pedagogical advisors, and academic support professionals can use the matrix as a framework to rethink their teaching and support strategies. In the “Learning Services” quadrant, it can inspire them to redesign learning experiences through service design principles like user-centeredness, iteration, and co-creation. Simultaneously, the matrix invites these actors to identify how support systems and infrastructures (e.g., IT tools, course management platforms, faculty development programs) influence pedagogical effectiveness, motivating systemic improvements across “Services for Learning.”

For administrative staff, the matrix acts as a map of institutional service flows, allowing them to identify how back-office systems (such as enrollment, IT support, library services, and housing) contribute to or hinder the educational experience. By understanding their role in the broader service system, they can become more proactive in aligning operational excellence with student-centered goals and institutional missions.

External public institutions (including policymakers and government agencies) may use the matrix to evaluate and guide policy formation and public investments. By highlighting the interdependence between internal structures and external outcomes, the matrix offers a lens to assess the systemic impact of education on social equity, innovation, and regional development. It can support more integrated policy design by exposing how educational services interact with wider societal systems. At the same time, private organizations and global educational bodies can apply the matrix to better understand how academic institutions generate value across different layers. It offers companies and networks a strategic interface for identifying synergies, designing collaborative programs, and co-investing in infrastructure, training, and applied research that reflect their shared interests with universities.

Finally, for communities and civil society stakeholders, the matrix reveals the institutional pathways for meaningful engagement. It makes visible how their contributions (whether

through participatory design labs or social innovation projects) can influence not only the university's outreach, but also its internal governance and educational agenda. In turn, communities gain access to relevant knowledge, platforms for expression, and tools to co-create sustainable futures.

By helping each stakeholder group situate themselves within a multidimensional service system, the matrix fosters a common language and a shared strategic vision. It supports inclusive dialogue, distributed innovation, and long-term alignment between institutional capabilities and societal challenges.

Discussion

The four quadrants proposed in this framework represent a complex service system for higher education, where each quadrant highlights a distinct aspect of the educational ecosystem. The first two quadrants are relatively straightforward to comprehend regarding the application of service design, as they directly engage with educational processes within the university. For instance, Learning Services can be illustrated by Calavia et al. (2023), who demonstrate how Design Thinking methodologies, based on the ideas of user-centricity, iterative prototyping, and collaborative co-creation, enable educators to become co-creators of learning experiences by designing instructional materials and approaches that are tailored to students' needs. Similarly, in the second quadrant, Services for Learning can be exemplified by Ostrom et al. (2011), who emphasize the use of service blueprints to optimize the internal university infrastructure, including administrative and operational support systems. On the other hand, the last two quadrants require a more systemic perspective and are less explored in current literature.

In recent years, universities have been facing increasing pressure to expand their focus, which previously centered primarily on teaching and research, to incorporate a Third Mission (TM) defined as "a contribution to society" (Compagnucci &Spigarelli, 2020). In this context, the fourth quadrant, the Educational Interaction Services quadrant, closely aligns with the TM of universities, offering significant opportunities for service design to facilitate collaboration and create a positive societal impact. Initiatives such as Off-Campus Nolo [1] and NICE 2035 [2] are innovative examples,

[1] A living lab initiated by Politecnico di Milano in the Nolo neighborhood of Milan. It serves as a platform for transferring academic knowledge and skills to the local community through a variety of activities, projects, and events. For more details, see https://doi.org/10.30682/diiddsi23t5v
[2] Launched in 2015, this project involves the College of Design and Innovation at Tongji University. It fosters a collaborative environment with research laboratories, educational spaces, and innovation

highlighting the potential of applying service design principles in the educational sector to foster meaningful partnerships between universities and local communities. Both initiatives incorporate fundamental service design principles, including human-centered design, where the needs and aspirations of community members are prioritized to ensure solutions are tailored to their specific contexts; value co-creation, with universities and communities collaborating to design and implement solutions together; prototyping and iteration, allowing solutions to be tested and refined in real-world environments for continuous learning and improvement; and stakeholders/actors collaboration, where students, faculty, local governments, and businesses work together, breaking down barriers and fostering diverse perspectives to create impactful solutions (Fassi & Vergani, 2024; Lou Yongqi, 2018).

In this way, we highlight that perceiving higher education as a complex service system beyond the academic realm is crucial for understanding its broader influence on economic development, innovation, and social justice (Lella et al., 2012; Morelli et al., 2021). This implies focusing on the network of relationships with various external stakeholders, including government entities, businesses, and local communities. The relevance of this systemic view becomes even more evident when considering national contexts such as India, where the coexistence of diverse institutional models, governance arrangements, and regional specificities reflects the presence of multiple interacting layers (Sheikh, 2017; Kapur, 2018). These layers give rise to heterogeneous and evolving configurations that underscore the adaptive, dynamic nature of educational systems. This perspective will highlight how political power dynamics and external influences shape academic autonomy and education choices.

On the other hand, the complexity of the relationship between the internal and external perspectives lies in how external forces shape internal practices and vice versa. For instance, external societal, cultural, and political factors influence how universities design and deliver services through different mechanisms, such as accreditation processes or collaborations between external stakeholders (companies, local institutions) and specific courses. At the same time, internal dynamics based on an institution's culture, values, and practices can influence how external policies are implemented and adapted. Therefore, this interconnectedness requires a holistic approach to recognizing the dynamic relationship between institutional culture, governance, and broader societal forces. By embracing this complexity, institutions can better manage change, improve educational outcomes, and contribute to creating a more equitable, effective, and responsive educational system.

centers, aiming to transform community dynamics and promote creativity and entrepreneurship among residents. For more details, see: https://mp.weixin.qq.com/s/luK1MIhg0Z4niedWaY-fQw

Based on the recognition of universities as complex service systems, we argue that service design can enhance their capacity to collaborate with stakeholders and create more effective educational services. Through its collaborative, practical, and iterative nature, service design can help universities identify societal needs, design solutions with real-world impact, and establish long-lasting partnerships that enhance academic learning and community development, especially in addressing the university's TM.

The distinction between "service design in education" and "service design for education" is also an important contribution to the field, as it helps delineate the scope of future studies. It provides a more structured understanding of how service design can be integrated into educational contexts, either as a specialized discipline or as a transversal approach that influences the design and delivery of education.

In this way, the positioning matrix proposed in this work can be a starting point for educators, administrators, policymakers, and service design professionals to apply service design concepts more effectively in educational contexts. By using the matrix, stakeholders can identify which areas of service design are relevant to their specific contexts, allowing them to align strategies and initiatives with their institutional goals. The matrix helps bridge the gap between theory and practice, providing a structured approach to understanding the complexities of educational systems.

Moreover, the framework offers a number of practical benefits that reinforce its potential applicability. It supports strategic alignment between institutional missions and service delivery, encourages participatory planning through the identification of internal and external stakeholders, and promotes interdisciplinary collaboration by connecting pedagogical, administrative, and societal dimensions. It also enhances the adaptive capacity of services, enabling institutions to respond to evolving needs, and can serve as a tool for policy development and evaluation, guiding the monitoring and continuous improvement of educational services.

Finally, while the application of service design for education can hold great perspective, further work is needed to explore the practical implementation of the concepts discussed. As this research employs a preliminary exploratory approach, future studies are needed to expand the findings. Future studies can focus on exploring deeper real cases that align with each of the four quadrants, providing more concrete examples of how service design can be effectively applied across different educational settings.

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